

Faculty Resume: Michael Caracotsios

1. Education

Ph.D. in Chemical Engineering - January 1986 at the **University of Wisconsin-Madison**

Thesis Advisor: Pr. Warren E. Stewart

Thesis Title: *Model Parametric Sensitivity Analysis and Nonlinear Parameter Estimation. Theory and Applications.*

Minor: Computer Science and Mathematics - GPA: 4.0/4.0

M.S. in Chemical Engineering - May, 1981 at the **University of Kentucky-Lexington**

GPA: 4.0/4.0 Rank # 1 in Graduating Class

Thesis Title: *Carbonylation of Methanol Catalyzed by Rhodium-Exchanged Zeolites.*

B.S. in Chemical Engineering - May, 1979 at the **University of Thessaloniki-Greece**

GPA: 9.3/10.0 Rank # 1 in Graduating Class

Thesis Title: *Estimation of the Alloying and Deoxidization Efficiency for 50tn Electric Furnace of a Steelmaking Plant in Northern Greece.*

2. Academic Experience

09/2015 to Present: Clinical Associate Professor- UIC IL.

Currently teaching undergraduate level courses in Chemical Engineering Thermodynamics, Chemical Process Control and Process Simulation.

01/2007 to 08/2015: Adjunct Professor- Northwestern University, Evanston IL.

Currently teaching undergraduate and graduate level courses in Computer Aided Process Design and Optimization, Applied Numerical Methods and Statistical Data Analysis in Chemical and Biological Engineering.

05/1986 to 12/2006: Adjunct Professor- Illinois Institute of Technology, Chicago IL.

Taught undergraduate and graduate level courses in Computer Aided Process Design and Optimization, Applied Numerical Methods and Statistical Data Analysis in Chemical Engineering, Fluid Mechanics and Heat Transfer. Received the Excellence in Teaching Award in the 1991-92 academic year.

3. Non-Academic Experience

08/1996 to 09/2015: *Research Fellow - UOP*, Optimization and Advanced Control Applications, Process Information and Control Division, Solutions and Services Division, Modeling and Optimization Division, Gas Processing Division.

03/1986 to 08/1996: *Senior Research Engineer - Amoco Chemical Company*, Research and Development Department, Aromatics Oxidation Division, Polymers Catalysis and Process Division, Computer Aided Process Engineering Division.

05/1984 to 09/1984: *Research Engineer, Chevron Oil Field Research Corporation*, Reservoir Engineering Division. Developed software on Cray supercomputers for steam-flood simulations in oil exploration.

4. Certifications or professional registrations

Green belt six-sigma certification

5. Current membership in professional organizations

American Institute of Chemical Engineers

6. Honors and awards

Honoring your Professor UIC Award (2017)
Honeywell Team Performance Award, for designing state of the art catalyst regeneration systems (2010)
BP Helios Award for developing state of the art optimization technology (2000)
William M. Langdon Award For Excellence in Teaching, IIT, Chicago (1991-1992)
First prize award, National Competition in Mathematics, Greece (1974)
Five awards for outstanding performance (ranked # 1) in Chemical Engineering from National Fellowship Foundation, Greece (1974-79).
Three awards for outstanding performance (ranked # 1) in the entire College of Engineering, from the Technical Chamber, Greece (1974-79).

7. Service activities

Freshmen Engineering Success Program Committee, participant, 2017-2019
Chemical Engineering Advisory Committee, August 2016-present
Presenter, High School District 203 Career Night, Engineering, Industrial Technology, College recruitment activities; (2014-2019)

8. Selected Publications

Kinetic Modelling of Water Inhibited Methane Oxidation Reactions over a Pd/Al₂O₃ Wash-coated Monolith using Spatially and Temporally Resolved Experimental Data, Ciaran Coney, Cristina Stere, Paul Millington, Agnes Raj, Sam Wilkinson, Geoffrey McCullough, Michael

Caracotsios, Chris Hardacre, Kevin Morgan, David Thompsett, Alexandre Goguet, ACS Catalysis (2020)

Nonlinear mixed-effects models for kinetic parameter estimation with batch reactor data, Daniel A. Hickman, Michael J. Ignatowich, Michael Caracotsios, James D. Sheehan and Fabio D'Ottaviano, *Chemical Engineering Journal* (2019)

Robustness of coke burning from catalyst for light paraffin dehydrogenation process, US Patent, US 8,895,464 B. Egolf, L. Leonard, M. Caracotsios (2014)

Computer Aided Modeling of Chemically Reactive Systems, co-authored with W. E. Stewart, *John Wiley and Sons, Inc.* , 2008

Innovations for refinery-wide modeling, simulation and optimization, Hydrocarbon Asia, 2001

Nontraditional optimization for isomerization unit improves profits, with T.C. Crowe *et al.*, Oil and Gas Journal, May 11, 1998

Polymerization of alpha-Olefins, (An advanced control method is disclosed for the polymerization of an alpha-olefin in a substantially horizontal, quench-cooled, stirred bed reactor) with A. Buchelli, *United States Patent*, Patent Number: 5,504,166, April 2, 1996.

Polymerization of alpha-Olefins, with A. Buchelli, *World Intellectual Property Organization*, International Bureau, International Publication Number WO 93/24533.

Sensitivity Analysis of Initial-Boundary-Value Problems with Mixed PDEs and Algebraic Equations. Applications to Chemical and Biochemical Systems, with W. E. Stewart, *Computers and Chemical Engineering*, Vol. 19, No. 9, pp. 1019-1030 (1995).

Sorption Kinetics and Axial Diffusion in Binary Solute Affinity Membrane Bioseparations, with M. R. Etzel, *Chemical Engineering Science*, Vol. 48, No. 10, pp. 1801-1812 (1993)

Theoretical Modeling of Amoco's Gas Phase Horizontal Stirred Bed Reactor for the Manufacturing of Polypropylene Resins, *Chemical Engineering Science*, Vol. 47, No. 9-11, pp. 2591-2596, 1992.

9. Professional Development

Industrial Multivariable Process Control

Computational Fluid Dynamics and Combinatorial Chemistry.

Real Time On-Line Optimization with ASPEN RT-OPT

Listening, Interpersonal and Communication Skills, Influencing without Direct Authority

Concepts in Quality Manufacturing and Technology Transfer

Management Principles and Techniques and Managing Personal Growth

Artificial Intelligence and Expert Systems in Engineering

Polymer Reaction Engineering, Poly-olefins Manufacture Featuring Metallocene Catalysts

Advance Modeling and Control of Polymerization Reactors
Applications of Pinch Technology to Chemical Industries
Principles of Heterogeneous Catalysis and Organo-metallic Chemistry
Advanced Programming with Microsoft Visual Basic 3.0 for Windows, C and PASCAL

10. Open-Source Textbook

Michael Caracotsios, *An Introduction to Chemical Engineering Thermodynamics. Theory and Applications*, Spring 2026